Package ‘GTAPViz’

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Title Automating GTAP Data Processing and Visualization

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Author Pattawee Puangchit <ppuangch@purdue.edu>

Description GTAPViz is an extension package of HARplus, designed to provide seamless data visualization for GTAP model results from `.HAR` and `.SL4` files. This package focuses on user-friendly visualization tools, enabling GTAP users to analyze and compare simulation results efficiently without requiring deep coding knowledge. With built-in functions for pivoting, mapping, unit assignments, and graphical representation, GTAPViz enhances economic model interpretation by generating structured, publication-ready plots. It is ideal for GTAP researchers, policymakers, and analysts looking for intuitive data exploration and visual insights from GEMPACK outputs.

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Encoding UTF-8

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RoxygenNote 7.3.2

BugReports https://github.com/bodysbobb/GTAPViz/issues

URL https://bodysbobb.github.io/GTAPViz/

Imports HARplus,
ggplot2,
dplyr,
tidyr,
openxlsx,
colorspace,
grDevices,
rlang,
scales,
purrr,
readxl,
writexl,
utils,
methods,
stringdist,
stats

Remotes github::Bodysbobb/HARplus

Suggests knitr,
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devtools

VignetteBuilder knitr

Depends R (>= 3.5)

Contents

add_mapping_info 2
auto_gtap_data 3
comparison_plot 6
convert_units 8
detail_plot 10
get_export_config 12
get_plot_config 13
get_plot_style_config 14
gtap_macros_data 17
rename_GTAP_bilateral 18
rename_value 19
report_table 20
stack_plot 21

Index **25**

`add_mapping_info` *Add Mapping Information to GTAP Data*

Description

Adds descriptions and unit information to GTAP data based on a specified mapping mode. Supports external mappings or default GTAPv7 mappings, allowing users to enrich datasets with standardized metadata.

Usage

```
add_mapping_info(
  data_list,
  external_map = NULL,
  mapping = "GTAPv7",
  description_info = TRUE,
  unit_info = TRUE
)
```

Arguments

`data_list`	A data structure containing GTAP variables.
`external_map`	Optional data frame with mapping information (must include "Variable", "Description", and "Unit" columns).
`mapping`	Character. Mapping mode: • `"GTAPv7"` (default): Uses standard GTAPv7 definitions. • `"Yes"`: Uses only the supplied descriptions and units from `external_map`. • `"No"`: Does not add any descriptions or units.

- "Mix": Prioritizes external_map, but falls back to GTAPv7 for missing values.

description_info
: Logical. If TRUE, adds description information to the data.

unit_info
: Logical. If TRUE, adds unit information to the data.

Value

A data structure with added mapping information, preserving the original structure.

Author(s)

Pattawee Puangchit

See Also

convert_units, rename_value

Examples

```
## Not run:
# Add mapping using GTAPv7 defaults
gtap_data <- add_mapping_info(gtap_data, mapping = "GTAPv7")

# Use an external mapping file
gtap_data <- add_mapping_info(gtap_data, external_map = my_mapping, mapping = "Mix")

## End(Not run)
```

---

auto_gtap_data *Process GTAP Data Automation with Flexible Output Options*

Description

Processes GTAP data from SL4 and HAR files with options for exporting and preparing plot-ready data.

Usage

```
auto_gtap_data(
  experiment,
  mapping_info = "GTAPv7",
  project_path = NULL,
  input_path = NULL,
  output_path = NULL,
  output_formats = NULL,
  plot_data = FALSE,
  region_select = NULL,
  sector_select = NULL,
  subtotal_level = FALSE,
  sl4_output_name = "sl4.plot.data",
```

```
  har_output_name = "har.plot.data",
  macro_output_name = "GTAPMacro",
  process_sl4_vars = NULL,
  process_har_vars = NULL,
  sl4_mapping_info = NULL,
  har_mapping_info = NULL,
  sl4_extract_method = "group_data_by_dims",
  har_extract_method = "get_data_by_var",
  sl4_priority = NULL,
  har_priority = NULL,
  sl4_suffix = "",
  har_suffix = "-WEL"
)
```

Arguments

- `experiment` Character vector. Case names to process.
- `mapping_info` Character. Mapping mode: "GTAPv7" (default), "Yes", "No", or "Mix".
- `project_path` Character. Path to the project folder with "in" and "out" subfolders.
- `input_path` Character. Path to the input folder. Overrides `project_path/in` if specified.
- `output_path` Character. Path to the output folder. Overrides `project_path/out` if specified.
- `output_formats` Character vector or list. Exports data in these formats (valid: "csv", "stata", "rds", "txt").
- `plot_data` Logical. If TRUE, prepares data for plotting and assigns to variables.
- `region_select` Optional character vector. Specifies regions to filter the data.
- `sector_select` Optional character vector. Specifies sectors to filter the data.
- `subtotal_level` Logical. If TRUE, includes subtotal data. Default is FALSE.
- `sl4_output_name` Character. Variable name for SL4 plotting data if generating plot data. Default is "sl4.plot.data".
- `har_output_name` Character. Variable name for HAR plotting data if generating plot data. Default is "har.plot.data".
- `macro_output_name` Character. Variable name for GTAP macro data if generating plot data. Default is "GTAPMacro".
- `process_sl4_vars` Data frame, NULL, or FALSE. Variables to extract from SL4 files. Set to NULL to extract all variables, or FALSE to skip SL4 processing.
- `process_har_vars` Data frame, NULL, or FALSE. Variables to extract from HAR files. Set to NULL to extract all variables, or FALSE to skip HAR processing.
- `sl4_mapping_info` Data frame or NULL. Mapping information for SL4 variables (with "Variable", "Description", and "Unit" columns).
- `har_mapping_info` Data frame or NULL. Mapping information for HAR variables (with "Variable", "Description", and "Unit" columns).

`sl4_extract_method`
: Character. SL4 extraction method. Options: "get_data_by_dims", "get_data_by_var", or "group_data_by_dims".

`har_extract_method`
: Character. HAR extraction method. Options: "get_data_by_dims", "get_data_by_var", or "group_data_by_dims".

`sl4_priority` Optional list. Priority rules for SL4 data grouping.

`har_priority` Optional list. Priority rules for HAR data grouping.

`sl4_suffix` Character. Custom suffix for SL4 files (e.g., "" or "-custom").

`har_suffix` Character. Custom suffix for HAR files (e.g., "-WEL").

Details

This function automates the workflow for processing GTAP model outputs, with flexible output options and optional filtering for region- and sector-specific data.

The key parameters `process_sl4_vars` and `process_har_vars` accept three different input types:

- **Data frame**: Contains variable mappings with required "Variable" column. When provided, only specified variables will be extracted.
- **NULL**: Extracts all available variables from the respective file type.
- **FALSE**: Completely skips processing of that file type, allowing the function to focus only on the other file type.

The `mapping_info` parameter controls how descriptions and units are assigned:

- **GTAPv7**: Uses standard GTAPv7 definitions (default).
- **Yes**: Uses only the supplied descriptions and units from `sl4_mapping_info`/`har_mapping_info`.
- **No**: Does not add any descriptions or units.
- **Mix**: Prioritizes supplied descriptions and units, falling back to GTAPv7 for any missing values.

Value

Returns the processed data invisibly, which will not be printed to the console.

Author(s)

Pattawee Puangchit

See Also

add_mapping_info, gtap_macros_data

Examples

```
## Not run:
# Extract data with region and experiment filters
auto_gtap_data(
  process_sl4_vars = NULL,
  process_har_vars = NULL,
  sl4_mapping_info = sl4_mapping_info,
  har_mapping_info = har_mapping_info,
```

```
  region_select = selected_regions,
  sector_select = NULL,
  subtotal_level = FALSE,
  experiment = experiment,
  mapping_info = "GTAPv7",
  project_path = project_folder,
  plot_data = TRUE,
  output_formats = list(
    "csv" = "No",
    "stata" = "No",
    "rds" = "No",
    "txt" = "No"))

## End(Not run)
```

`comparison_plot` *Create Comparative Bar Charts from HAR and SL4 Data*

Description

Generates comparative bar charts using GTAP data, allowing multiple visualization options such as panel facets, split grouping, and customizable styles.

Usage

```
comparison_plot(
  data,
  filter_var = NULL,
  x_axis_from,
  split_by = NULL,
  panel_var = "Experiment",
  variable_col = "Variable",
  unit_col = "Unit",
  desc_col = "Description",
  invert_pane = FALSE,
  separate_figure = FALSE,
  var_name_by_description = FALSE,
  add_var_info = FALSE,
  output_path = NULL,
  export_picture = TRUE,
  export_as_pdf = FALSE,
  export_config = NULL,
  plot_style_config = NULL
)
```

Arguments

`data`	A data frame or a list of data frames containing GTAP results.
`filter_var`	Vector or data frame. If a vector, filters the values in `x_axis_from`. If a data frame, filters `value_col` based on matching `variable_col` values.
`x_axis_from`	Character. Column name for x-axis categories (e.g., "REG", "Sector").

`split_by`	Character or vector. Column name(s) for data splitting (e.g., "COMM", "REG", "ACTS"). Set to NULL for no splitting, which is suitable for macro-level analysis (i.e., aggregated values without additional dimensions).
`panel_var`	Character. Column for panel facets (default: "Experiment").
`variable_col`	Character. Column containing variable identifiers (default: "Variable").
`unit_col`	Character. Column containing unit information (default: "Unit").
`desc_col`	Character. Column containing variable descriptions (default: "Description").
`invert_pane`	Logical. If TRUE, creates horizontal bars instead of vertical ones (default: FALSE).
`separate_figure`	Logical. If TRUE, generates separate figures per panel value (default: FALSE).
`var_name_by_description`	Logical. If TRUE, uses descriptions instead of variable codes in titles (default: FALSE).
`add_var_info`	Logical. If TRUE, adds the variable code in parentheses after the description (default: FALSE).
`output_path`	Character. Directory path for saving output files. If NULL, plots are only returned in R.
`export_picture`	Logical. If TRUE, exports plots as image files (default: TRUE).
`export_as_pdf`	Logical. If TRUE, exports plots as PDF files instead of PNG (default: FALSE).
`export_config`	List. Configuration options for exported plots, including file name, width, and height.
`plot_style_config`	List. Custom style configuration for plots. If NULL, defaults from `get_plot_style_config("compa` are applied.

Details

This function allows for extensive customization of plots through the `plot_style_config` parameter, which integrates ggplot2 configurations. The function now supports more flexible export options with the `export_picture`, `export_as_pdf`, and `export_config` parameters.

The `export_config` parameter can include:

- `file_name`: Base name for the output files (default: "comparison_plots").
- `width`: Width of the output files in inches.
- `height`: Height of the output files in inches.
- Additional export settings as needed.

Value

A ggplot2 object for a single plot or a list of ggplot2 objects for multiple plots.

Author(s)

Pattawee Puangchit

See Also

`detail_plot`, `stack_plot`, `get_plot_style_config`

Examples

```
## Not run:
# Basic usage with data frame
p1 <- comparison_plot(
  data = gtap_results,
  x_axis_from = "REG",
  panel_var = "Experiment"
)

# Split by commodity with custom styling and export options
p2 <- comparison_plot(
  data = gtap_results,
  x_axis_from = "REG",
  split_by = "COMM",
  panel_var = "Experiment",
  var_name_by_description = TRUE,
  output_path = "results/plots",
  export_as_pdf = TRUE,
  export_config = list(
    file_name = "commodity_impacts",
    width = 12,
    height = 8
  ),
  plot_style_config = list(
    color_tone = "economic",
    title_size = 16,
    show_grid_major_y = TRUE
  )
)

## End(Not run)
```

`convert_units` *Convert Units in GTAP Data*

Description

Converts values in a dataset to different units based on predefined transformations or custom scaling. Supports manual and automatic conversions for economic and trade-related metrics.

Usage

```
convert_units(
  data,
  change_unit_from = NULL,
  change_unit_to = NULL,
  adjustment = NULL,
  value_col = "Value",
  unit_col = "Unit",
  variable_select = NULL,
  variable_col = "Variable",
  scale_auto = NULL
)
```

Arguments

- `data` — A data structure (list, data frame, or nested combination).
- `change_unit_from` — Character vector. Units to be converted (case-insensitive).
- `change_unit_to` — Character vector. Target units corresponding to `change_unit_from`.
- `adjustment` — Character or numeric vector. Specifies conversion operations (e.g., `"/1000"` to convert million to billion).
- `value_col` — Character. Column name containing values to adjust (default: `"Value"`).
- `unit_col` — Character. Column name containing unit information (default: `"Unit"`).
- `variable_select` — Optional character vector. If provided, only these variables are converted.
- `variable_col` — Character. Column name containing variable identifiers (default: `"Variable"`).
- `scale_auto` — Optional character vector of predefined conversion rules:
  - `"mil2bil"`: Converts million USD to billion USD (divides by 1000).
  - `"bil2mil"`: Converts billion USD to million USD (multiplies by 1000).
  - `"pct2frac"`: Converts percent to fraction (divides by 100).
  - `"frac2pct"`: Converts fraction to percent (multiplies by 100).

Details

If both `change_unit_from` and `scale_auto` are provided, the function prompts the user to choose between manual and automatic conversion.

Value

A data structure with values converted to the specified units.

Author(s)

Pattawee Puangchit

See Also

`add_mapping_info`, `rename_value`

Examples

```
## Not run:
# Convert million USD to billion USD
gtap_data <- convert_units(gtap_data,
  change_unit_from = "million USD",
  change_unit_to = "billion USD",
  adjustment = "/1000"
)

# Automatic conversion from percent to fraction
gtap_data <- convert_units(gtap_data, scale_auto = "pct2frac")

## End(Not run)
```

detail_plot *Create Comprehensive Detailed Bar Charts for Impact Distribution*

Description

Generates detailed bar charts to visualize the distribution of impacts across multiple dimensions, automatically handling multi-dimensional data. The function supports top impact filtering, color coding for positive and negative values, and flexible visualization settings.

Usage

```
detail_plot(
  data,
  filter_var = NULL,
  x_axis_from,
  split_by = NULL,
  panel_var = "Experiment",
  variable_col = "Variable",
  unit_col = "Unit",
  desc_col = "Description",
  var_name_by_description = FALSE,
  add_var_info = FALSE,
  output_path = NULL,
  export_picture = TRUE,
  export_as_pdf = FALSE,
  export_config = NULL,
  top_impact = NULL,
  separate_figure = FALSE,
  invert_pane = FALSE,
  plot_style_config = NULL
)
```

Arguments

- `data` A data frame or a list of data frames containing GTAP results.
- `filter_var` Vector or data frame. If a vector, filters the values in `x_axis_from`. If a data frame, filters `value_col` based on matching `variable_col` values.
- `x_axis_from` Character. Column name for x-axis categories (e.g., "REG", "Sector").
- `split_by` Character or vector. Column name(s) for data splitting (e.g., "COMM", "REG", "ACTS"). Set to NULL for no splitting, which is suitable for macro-level analysis (i.e., aggregated values without additional dimensions).
- `panel_var` Character. Column for panel facets (default: "Experiment").
- `variable_col` Character. Column containing variable identifiers (default: "Variable").
- `unit_col` Character. Column containing unit information (default: "Unit").
- `desc_col` Character. Column containing variable descriptions (default: "Description").
- `var_name_by_description` Logical. If TRUE, uses descriptions instead of variable codes in titles (default: FALSE).

- `add_var_info` — Logical. If TRUE, adds the variable code in parentheses after the description (default: FALSE).
- `output_path` — Character. Directory path for saving output files. If NULL, plots are only returned in R.
- `export_picture` — Logical. If TRUE, exports plots as image files (default: TRUE).
- `export_as_pdf` — Logical. If TRUE, exports plots as PDF files instead of PNG (default: FALSE).
- `export_config` — List. Configuration options for exported plots, including file name, width, and height.
- `top_impact` — Numeric or NULL. If specified, filters to show only the top N impactful values. NULL shows all values.
- `separate_figure` — Logical. If TRUE, creates a separate figure for each panel value (default: FALSE).
- `invert_pane` — Logical. If TRUE, creates horizontal bars instead of vertical ones (default: FALSE).
- `plot_style_config` — List. Custom style configuration for plots. If NULL, defaults from `get_plot_style_config("detai` are applied.

Details

This function allows for extensive customization of plots through the `plot_style_config` parameter, which integrates ggplot2 configurations. The function now supports more flexible export options with the `export_picture`, `export_as_pdf`, and `export_config` parameters.

The `export_config` parameter can include:

- `file_name`: Base name for the output files (default: "detail_plots").
- `width`: Width of the output files in inches.
- `height`: Height of the output files in inches.
- Additional export settings as needed.

Value

A ggplot2 object for a single plot or a list of ggplot2 objects for multiple plots.

Author(s)

Pattawee Puangchit

See Also

`comparison_plot`, `stack_plot`, `get_plot_style_config`

Examples

```
## Not run:
# Basic usage showing all impacts with export options
p1 <- detail_plot(
  data = gtap_results,
  x_axis_from = "REG",
  variable_col = "Variable",
  top_impact = NULL,
```

```
  output_path = "results/detail_plots",
  export_config = list(
    file_name = "regional_impacts",
    width = 14,
    height = 10
  )
)

# Show only top 10 impacts (balanced between positive and negative)
p2 <- detail_plot(
  data = gtap_results,
  x_axis_from = "Sector",
  split_by = "Region",
  panel_var = "Experiment",
  variable_col = "Variable",
  top_impact = 10,
  invert_pane = TRUE,
  export_as_pdf = TRUE
)

## End(Not run)
```

`get_export_config` *Get Export Configuration Options*

Description

Returns documentation and default values for export configuration options used in plotting functions.

Usage

```
get_export_config(as_dataframe = FALSE)
```

Arguments

`as_dataframe`	Logical. Whether to return settings as a dataframe. Default is FALSE.

Details

Export Configuration Parameters:

- `file_name`: Character. Base name for exported files. Default: `"gtap_plots"`
- `width`: Numeric or `NULL`. Plot width in inches. Default: `NULL` (automatically calculated)
- `height`: Numeric or `NULL`. Plot height in inches. Default: `NULL` (automatically calculated)
- `dpi`: Numeric. Resolution for PNG export. Default: `300`
- `bg`: Character. Background color. Default: `"white"`
- `limitsize`: Logical. Whether to limit size. Default: `FALSE`

Value

If `as_dataframe` is TRUE, returns a dataframe with export configuration settings. Otherwise, returns a list with configuration settings.

Author(s)

Pattawee Puangchit

See Also

comparison_plot, detail_plot, stack_plot

Examples

```
# Get export configuration as list
export_info <- get_export_config()

# Get as a formatted dataframe
export_df <- get_export_config(as_dataframe = TRUE)
```

---

get_plot_config   *Get Plot and Export Configurations*

Description

Returns a list containing plot style configuration and export configuration. The function supports retrieving a single plot style or all available styles.

Usage

```
get_plot_config(plot_style = "all", config = NULL, export_config = NULL)
```

Arguments

`plot_style`	Character. The plot style to retrieve: "comparison", "detail", "stack", or "all". Default is "all", which returns configurations for all styles.
`config`	List. Optional custom style configuration to override defaults.
`export_config`	List. Optional custom export configuration to override defaults.

Value

A list with two components:

- `plot_style_config`: Plot style configuration(s). If `plot_style="all"`, contains a nested list with configurations for all styles.
- `export_config`: Export configuration as a data frame.

Author(s)

Pattawee Puangchit

See Also

comparison_plot, detail_plot, stack_plot

Examples

```
# Get all plot configurations with default settings
all_configs <- get_plot_config()

# Get only comparison plot configuration
comp_config <- get_plot_config("comparison")
```

---

get_plot_style_config   *Get Plot Style Configuration*

Description

Returns configuration settings for plot styles, with options to view as a structured dataframe or to look up specific parameters. Also provides parameter validation for custom configurations.

Usage

```
get_plot_style_config(
  plot_type = "comparison",
  parameter_name = NULL,
  show_docs = FALSE,
  validate_custom = NULL,
  as_dataframe = FALSE
)
```

Arguments

`plot_type`	Character. Type of plot: "comparison" (default), "detail", or "stack".
`parameter_name`	Character or NULL. Name of specific parameter to return information about.
`show_docs`	Logical. Whether to include documentation in the output.
`validate_custom`	List or NULL. Custom configuration settings to validate.
`as_dataframe`	Logical. Whether to return settings as a dataframe.

Details

This function applies default styling for different types of plots and allows users to customize the appearance. The parameters are grouped as follows:

Title Settings:

- `show_title`: Logical. Show or hide the plot title. Default: `TRUE`
- `title_face`: Character. Font face (`"bold"`, `"plain"`, `"italic"`). Default: `"bold"`
- `title_size`: Numeric. Font size of title. Default: `20`
- `title_hjust`: Numeric. Horizontal alignment (0 = left, 1 = right). Default: `0.5`
- `add_unit_to_title`: Logical. Append unit to title if applicable. Default: `TRUE`
- `title_margin`: ggplot2 `margin()` object. Default: `ggplot2::margin(10, 0, 10, 0)`
- `title_format`: List. Defines title formatting options (`type = "standard"`, `text = ""`)

X-Axis Settings:

- show_x_axis_title: Logical. Show or hide x-axis title. Default: FALSE
- x_axis_title_face: Character. Font face for x-axis title. Default: "bold"
- x_axis_title_size: Numeric. Font size of x-axis title. Default: 16
- x_axis_title_margin: ggplot2 margin(). Default: ggplot2::margin(t = 20)
- show_x_axis_labels: Logical. Show or hide x-axis labels. Default: TRUE
- x_axis_text_face: Character. Font face for x-axis labels. Default: "bold"
- x_axis_text_size: Numeric. Font size of x-axis labels. Default: 14
- x_axis_text_angle: Numeric. Angle of x-axis labels. Default: 45
- x_axis_text_hjust: Numeric. Horizontal justification of x-axis labels. Default: 1
- x_axis_description: Character. Optional description for the x-axis. Default: ""

Y-Axis Settings:

- show_y_axis_title: Logical. Show or hide y-axis title. Default: TRUE
- y_axis_title_face: Character. Font face for y-axis title. Default: "bold"
- y_axis_title_size: Numeric. Font size of y-axis title. Default: 16
- y_axis_title_margin: ggplot2 margin(). Default: ggplot2::margin(r = 20)
- show_y_axis_labels: Logical. Show or hide y-axis labels. Default: TRUE
- y_axis_text_face: Character. Font face for y-axis labels. Default: "plain"
- y_axis_text_size: Numeric. Font size of y-axis labels. Default: 14
- y_axis_text_angle: Numeric. Angle of y-axis labels. Default: 0
- y_axis_text_hjust: Numeric. Horizontal justification of y-axis labels. Default: 0
- y_axis_description: Character. Optional description for the y-axis. Default: ""
- show_axis_titles_on_all_facets: Logical. Show axis titles on all facets. Default: TRUE

Value Label Settings:

- show_value_labels: Logical. Show or hide value labels. Default: TRUE
- value_label_face: Character. Font face for value labels. Default: "plain"
- value_label_size: Numeric. Font size of value labels. Default: 5
- value_label_position: Character. Position of value labels ("above", "outside", "top"). Default: "above"
- value_label_decimal_places: Numeric. Number of decimal places in value labels. Default: 2

Legend Settings:

- show_legend: Logical. Show or hide legend. Default: FALSE
- show_legend_title: Logical. Show or hide legend title. Default: FALSE
- legend_position: Character. Legend position ("none", "bottom", "right"). Default: "none"
- legend_title_face: Character. Font face for legend title. Default: "bold"
- legend_text_face: Character. Font face for legend text. Default: "plain"
- legend_text_size: Numeric. Font size for legend text. Default: 14

Panel Strip Settings:

- strip_face: Character. Font face for panel strip. Default: "bold"
- strip_text_size: Numeric. Font size for panel strip. Default: 16
- strip_background: Character. Background color of strip. Default: "lightgrey"
- strip_text_margin: ggplot2 margin(). Default: ggplot2::margin(10, 0, 10, 0)

Panel Layout:

- panel_spacing: Numeric. Spacing between panels. Default: 2
- panel_rows: Numeric or NULL. Number of rows in panel layout. Default: NULL
- panel_cols: Numeric or NULL. Number of columns in panel layout. Default: NULL
- theme: ggplot2 theme object or NULL. Custom ggplot theme. Default: NULL

Color Settings:

- color_tone: Character or NULL. Base color theme. Default: NULL
- positive_color: Character. Color for positive values. Default: "#2E8B57"
- negative_color: Character. Color for negative values. Default: "#CD5C5C"
- background_color: Character. Background color of plot. Default: "white"
- grid_color: Character. Color of grid lines. Default: "grey90"
- show_grid_major_x: Logical. Show major grid lines on x-axis. Default: FALSE
- show_grid_major_y: Logical. Show major grid lines on y-axis. Default: TRUE
- show_grid_minor_x: Logical. Show minor grid lines on x-axis. Default: FALSE
- show_grid_minor_y: Logical. Show minor grid lines on y-axis. Default: FALSE

Zero Line Settings:

- show_zero_line: Logical. Show or hide zero line. Default: TRUE
- zero_line_type: Character. Line type ("solid", "dashed", "dotted"). Default: "dashed"
- zero_line_color: Character. Color of zero line. Default: "black"
- zero_line_size: Numeric. Line thickness of zero line. Default: 0.5
- zero_line_position: Numeric. Position of the zero line. Default: 0

Bar Chart Settings:

- bar_width: Numeric. Width of bars. Default: 0.9
- bar_spacing: Numeric. Spacing between bars. Default: 0.9

Scale Settings:

- scale_limit: Numeric vector of length 2. Manual limits for value axis. Example: c(-10, 10)
- scale_increment: Numeric. Step size for axis tick marks. Example: 2

Scale Expansion Settings:

- expansion_y_mult: Numeric vector. Y-axis expansion. Default: c(0.05, 0.1)
- expansion_x_mult: Numeric vector. X-axis expansion. Default: c(0.05, 0.05)

All Font Adjustment:

- all_font_size: Numeric. Master control for all font sizes. Default: 1

Value

If parameter_name is specified, returns a list with the value and documentation for that parameter. If validate_custom is specified, returns the validated configuration list. If as_dataframe is TRUE, returns a dataframe with configuration settings. Otherwise, returns a list with all configuration settings.

Author(s)

Pattawee Puangchit

See Also

comparison_plot, detail_plot, stack_plot

Examples

```
# Get default configuration
config <- get_plot_style_config("comparison")

# Get information about a specific parameter
param_info <- get_plot_style_config("detail", "bar_width")

# Get as structured dataframe
config_df <- get_plot_style_config("stack", as_dataframe = TRUE)

# Validate custom configuration
custom_config <- list(title_size = 24, bar_width = 0.7)
validated <- get_plot_style_config("comparison", validate_custom = custom_config)
```

---

gtap_macros_data — *Extract and Aggregate Scalar Macroeconomic Variables*

Description

Extracts scalar macroeconomic variables from multiple SL4 datasets and aggregates them into a structured data frame.

Usage

```
gtap_macros_data(
  input_dir = NULL,
  experiment_names = NULL,
  output_dir = NULL,
  output_formats = NULL,
  subtotal_level = FALSE,
  select_var = NULL
)
```

Arguments

input_dir	Character. Directory containing SL4 files.
experiment_names	Character vector. List of experiment names corresponding to SL4 files.
output_dir	Character (optional). Directory to save exported data.
output_formats	Character vector (optional). List of output formats (e.g., "csv", "xlsx").
subtotal_level	Logical. Whether to include subtotal levels in the processed data.
select_var	Character vector (optional). List of specific variable names to filter from the final result. If NULL, all variables are returned.

Value

A sorted dataframe containing processed GTAP macro data.

Author(s)

Pattawee Puangchit

See Also

add_mapping_info, process_gtap_data

Examples

```
## Not run:
# Method 1: Extract all variables
input_dir <- paste0(input.folder)
experiment <- c("EXP1", "EXP2") # File name (.sl4)
Macros <- gtap_macros_data(input_dir, experiment_names = experiment,
subtotal_level = FALSE)

# Method 2: Filter specific variables
Macros <- gtap_macros_data(input_dir, experiment_names = experiment,
                          select_var = c("qgdp", "pop", "gdpexp"))

## End(Not run)
```

---

rename_GTAP_bilateral *Rename GTAP Bilateral Trade Columns*

Description

Renames bilateral trade columns in GTAP data to standardized names, ensuring consistency in regional trade flows.

Usage

```
rename_GTAP_bilateral(data)
```

Arguments

data	Data structure containing GTAP bilateral trade data.

Value

The same data structure with renamed bilateral trade columns.

Author(s)

Pattawee Puangchit

See Also

add_mapping_info, convert_units

Examples

```
## Not run:
# Rename bilateral trade columns in a GTAP dataset
gtap_data <- rename_GTAP_bilateral(gtap_data)

## End(Not run)
```

---

rename_value *Rename Values in a Column*

Description

Replaces specific values in a column based on a provided mapping file. Supports renaming across nested data structures and preserves factor levels.

Usage

```
rename_value(data, column_name = NULL, mapping.file)
```

Arguments

data	Data structure (data frame, list, or nested combination).
column_name	Character. Column to modify. If NULL, the function extracts it from mapping.file.
mapping.file	Data frame with "OldName" and "NewName" columns for renaming.

Value

The same data structure with specified values replaced.

Author(s)

Pattawee Puangchit

See Also

add_mapping_info, convert_units

Examples

```
## Not run:
# Rename variables in a dataset
rename_map <- data.frame(OldName = c("old_var1", "old_var2"), NewName = c("new_var1", "new_var2"))
gtap_data <- rename_value(gtap_data, column_name = "Variable", mapping.file = rename_map)

## End(Not run)
```

report_table *Generate a Detailed Report Table*

Description

Transforms one or more datasets into a wide-format table based on specified grouping columns and pivot column(s). Supports subtotal filtering, renaming, and exporting to Excel.

Usage

```
report_table(
  data_list,
  pivot_col,
  total_column = FALSE,
  export_table = FALSE,
  separate_file = FALSE,
  output_dir = NULL,
  sheet_names = NULL,
  include_units = FALSE,
  component_exclude = NULL,
  group_by = NULL,
  rename_cols = NULL,
  var_name_by_description = TRUE,
  add_var_info = FALSE,
  decimal = 2,
  unit_select = NULL,
  separate_sheet_by = NULL,
  subtotal_level = FALSE,
  repeat_label = FALSE,
  workbook_name = "detail_results",
  add_group_line = FALSE
)
```

Arguments

`data_list`	A named list of data frames to be processed.
`pivot_col`	A named list specifying which column is pivoted into wide format for each dataset.
`total_column`	Logical. If `TRUE`, includes a "Total" column summing numeric values.
`export_table`	Logical. If `TRUE`, exports the tables to an Excel file.
`separate_file`	Logical. If `TRUE`, saves each dataset in a separate Excel file.
`output_dir`	Character. Directory where Excel files are written if `export_table` = `TRUE`.
`sheet_names`	Optional named list for custom sheet names.
`include_units`	Logical. If `TRUE`, includes the "Unit" column in grouping.
`component_exclude`	Optional character vector specifying pivoted values to exclude.
`group_by`	A list or character vector of grouping columns, or `NULL` for automatic detection.
`rename_cols`	A named list for renaming columns.

var_name_by_description
: Logical. If TRUE, replaces Variable names with their Description.

add_var_info
: Logical. If TRUE, appends the Variable code in parentheses after the Description.

decimal
: Numeric. Number of decimal places for rounding values.

unit_select
: Optional. Filters rows based on the specified unit.

separate_sheet_by
: Optional column name to split sheets in Excel.

subtotal_level
: Logical. If TRUE, includes all subtotal values; otherwise, keeps only Subtotal = "TOTAL".

repeat_label
: Logical. If TRUE, repeats the first group column in exports for clarity.

workbook_name
: Character. Name of the Excel workbook (without extension).

add_group_line
: Logical. If TRUE, adds a thin line after each group in the exported table.

Value

A named list of transformed data frames. If export_table = TRUE, tables are saved as Excel files.

Author(s)

Pattawee Puangchit

See Also

add_mapping_info, convert_units, rename_value

Examples

```
## Not run:
# Create a detailed report table with wide-format pivoting
report_data <- report_table(
  data_list = my_data,
  pivot_col = list("Variable" = "Value"),
  export_table = TRUE,
  output_dir = "results/"
)

## End(Not run)
```

---

stack_plot — *Create Comprehensive Stacked Bar Charts for Decomposition Analysis*

Description

Generates stacked bar charts to visualize the composition of values across multiple dimensions, automatically handling multi-dimensional data. The function supports both stacked and unstacked presentations, making it particularly useful for decomposition analysis.

Usage

```
stack_plot(
  data,
  filter_var = NULL,
  x_axis_from,
  stack_value_from,
  split_by = NULL,
  panel_var = "Experiment",
  variable_col = "Variable",
  unit_col = "Unit",
  desc_col = "Description",
  invert_pane = FALSE,
  separate_figure = FALSE,
  unstack_plot = FALSE,
  output_path = NULL,
  export_picture = TRUE,
  export_as_pdf = FALSE,
  export_config = NULL,
  var_name_by_description = FALSE,
  add_var_info = FALSE,
  show_total = TRUE,
  top_impact = NULL,
  plot_style_config = NULL
)
```

Arguments

`data`	A data frame or a list of data frames containing GTAP results.
`filter_var`	Vector or data frame. If a vector, filters the values in `x_axis_from`. If a data frame, filters `value_col` based on matching `variable_col` values.
`x_axis_from`	Character. Name of the column to use for x-axis categories (e.g., "REG", "Sector").
`stack_value_from`	Character. Name of the column containing stack component categories (e.g., "COMM" for commodities).
`split_by`	Character or vector. Column name(s) for data splitting (e.g., "COMM", "REG", "ACTS"). Set to NULL for no splitting, which is suitable for macro-level analysis (i.e., aggregated values without additional dimensions).
`panel_var`	Character. Column for panel facets (default: "Experiment").
`variable_col`	Character. Column containing variable identifiers (default: "Variable").
`unit_col`	Character. Column containing unit information (default: "Unit").
`desc_col`	Character. Column containing variable descriptions (default: "Description").
`invert_pane`	Logical. If TRUE, creates horizontal bars instead of vertical ones (default: FALSE).
`separate_figure`	Logical. If TRUE, creates a separate figure for each panel value (default: FALSE).
`unstack_plot`	Logical. If TRUE, creates separate bar plots for each `x_axis_from` value instead of stacked bars (default: FALSE).

`output_path`	Character. Directory path for saving output files. If NULL, plots are only returned in R.
`export_picture`	Logical. If TRUE, exports plots as image files (default: TRUE).
`export_as_pdf`	Logical. If TRUE, exports plots as PDF files instead of PNG (default: FALSE).
`export_config`	List. Configuration options for exported plots, including file name, width, and height.
`var_name_by_description`	Logical. If TRUE, uses descriptions instead of variable codes in titles (default: FALSE).
`add_var_info`	Logical. If TRUE, adds the variable code in parentheses after the description (default: FALSE).
`show_total`	Logical. If TRUE, displays total values above stacked bars (default: TRUE).
`top_impact`	Numeric or NULL. If specified, filters to show only the top N impactful values. NULL shows all values.
`plot_style_config`	List. Custom style configuration for plots. If NULL, defaults from `get_plot_style_config("stack` are applied.

Details

This function allows for extensive customization of plots through the `plot_style_config` parameter, which integrates ggplot2 configurations. The function now supports more flexible export options with the `export_picture`, `export_as_pdf`, and `export_config` parameters.

The `export_config` parameter can include:

- `file_name`: Base name for the output files (default: "stack_plots").
- `width`: Width of the output files in inches.
- `height`: Height of the output files in inches.
- Additional export settings as needed.

Value

A ggplot2 object for a single plot or a list of ggplot2 objects for multiple plots.

Author(s)

Pattawee Puangchit

See Also

`detail_plot`, `comparison_plot`, `get_plot_style_config`

Examples

```
## Not run:
# Basic stacked bar chart by commodity with export options
p1 <- stack_plot(
  data = gtap_results,
  x_axis_from = "REG",
  stack_value_from = "COMM",
  output_path = "results/stacked",
```

```
  export_config = list(
    file_name = "commodity_stacks",
    width = 12,
    height = 9
  )
)

# Welfare Decomposition with PDF export
p2 <- stack_plot(
  data = headerA,
  x_axis_from = "Region",
  stack_value_from = "COLUMN",
  split_by = FALSE,
  unstack_plot = TRUE,
  show_total = TRUE,
  export_as_pdf = TRUE,
  export_config = list(
    file_name = "welfare_decomposition"
  )
)

## End(Not run)
```

Index

add_mapping_info, 2, *5*, *9*, *18*, *19*, *21*
auto_gtap_data, 3

comparison_plot, 6, *11*, *13*, *17*, *23*
convert_units, *3*, 8, *18*, *19*, *21*

detail_plot, *7*, 10, *13*, *17*, *23*

get_export_config, 12
get_plot_config, 13
get_plot_style_config, *7*, *11*, 14, *23*
gtap_macros_data, *5*, 17

process_gtap_data, *18*

rename_GTAP_bilateral, 18
rename_value, *3*, *9*, 19, *21*
report_table, 20

stack_plot, *7*, *11*, *13*, *17*, 21